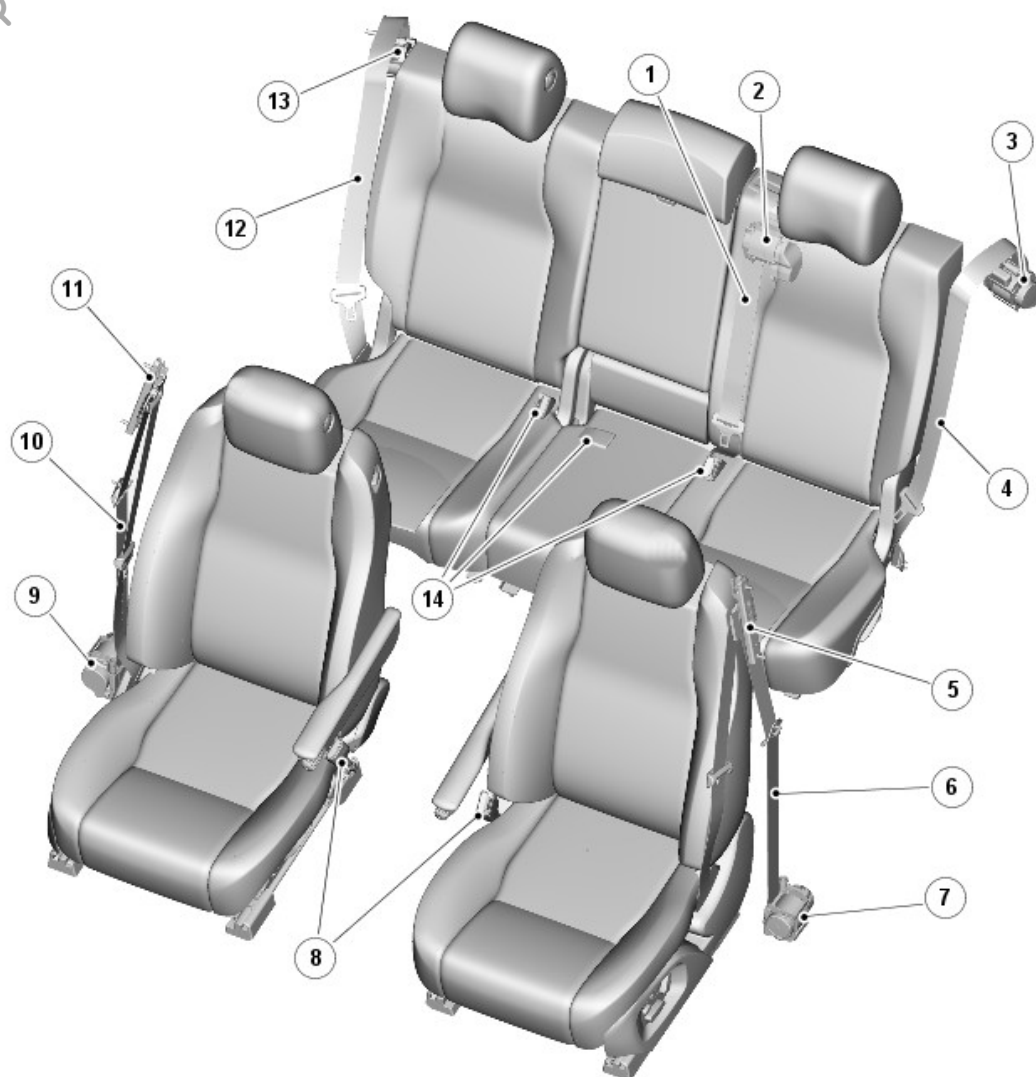


2016.0 RANGE ROVER (LG), 501-20

SEATBELT SYSTEM

DESCRIPTION AND OPERATION

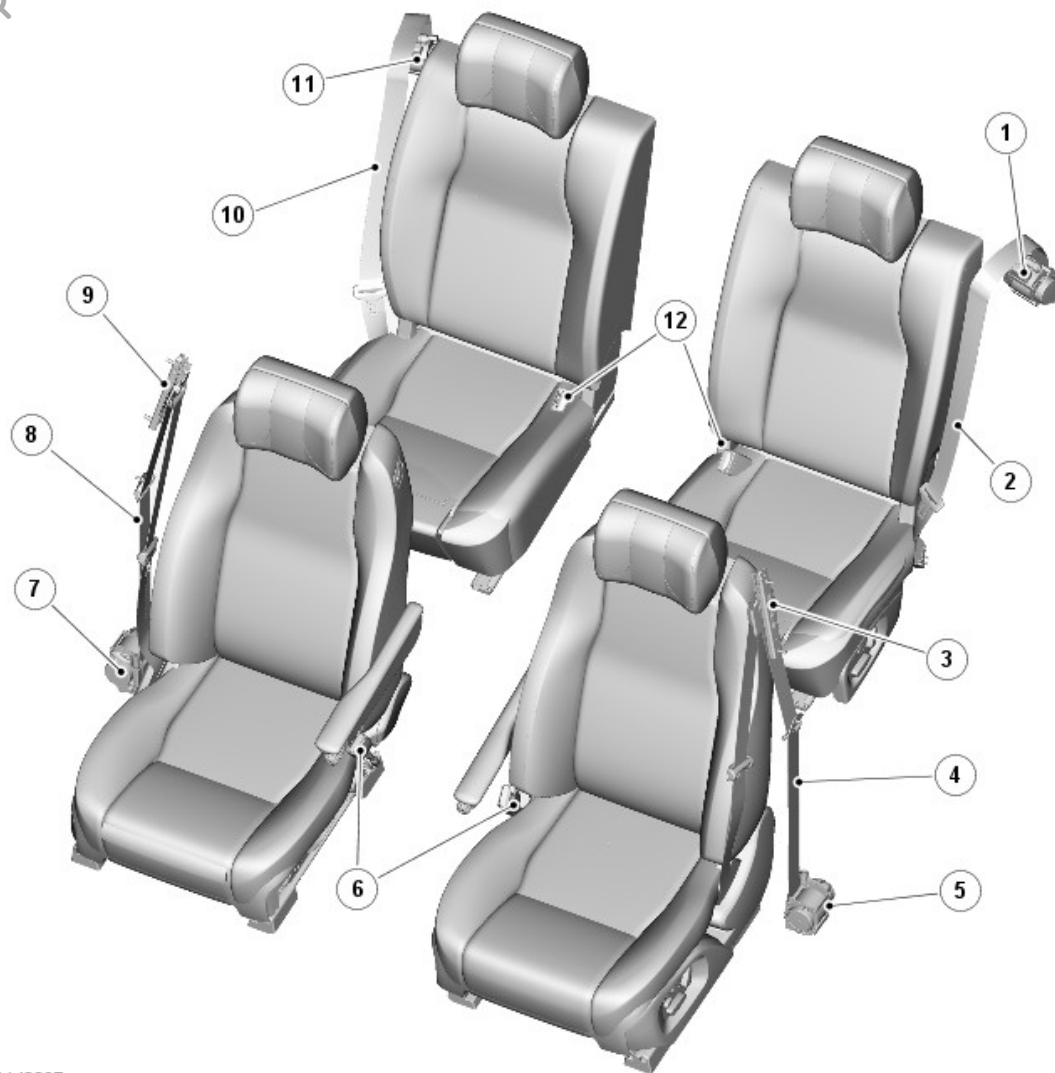
COMPONENT LOCATION - 5 SEAT VEHICLES



E142396

ITEM	DESCRIPTION
1	Center rear seatbelt
2	Center rear seatbelt retractor
3	Left rear seatbelt retractor
4	Left rear seatbelt
5	Left front seatbelt height adjuster
6	Left front seatbelt
7	Left front seatbelt retractor
8	Front seatbelt buckles
9	Right front seatbelt retractor
10	Right front seatbelt
11	Right front seatbelt height adjuster
12	Right rear seatbelt
13	Right rear seatbelt retractor
14	Rear seatbelt buckles

COMPONENT LOCATION - 4 SEAT VEHICLES



E142397

ITEM	DESCRIPTION
1	Left rear seatbelt retractor
2	Left rear seatbelt
3	Left front seatbelt height adjuster
4	Left front seatbelt
5	Left front seatbelt retractor
6	Front seatbelt buckles
7	Right front seatbelt retractor
8	Right front seatbelt
9	Right front seatbelt height adjuster
10	Right rear seatbelt

11	Right rear seatbelt retractor
12	Rear seatbelt buckles

OVERVIEW

A three point seatbelt is installed at each seat position. The front seatbelts have automatic height adjusters installed on the B pillars.

A pretensioner is incorporated into the retractor of each front seatbelt and of the two rear seatbelts (4 seat vehicles), or the two outer rear seatbelts (5 seat vehicles). Depending on vehicle specification and market, the seatbelts may also have:

- A tension sensor incorporated into the anchor of the front passenger seatbelt.
- An ALR (automatic locking retractor) fitted to all passenger carrying seats.
- A load limiter actuator incorporated into the retractor of the front passenger seatbelt.
- A pretensioner incorporated into the anchor of the driver seatbelt.
- A retractor with an active seatbelt function, on the driver and front passenger seatbelts.

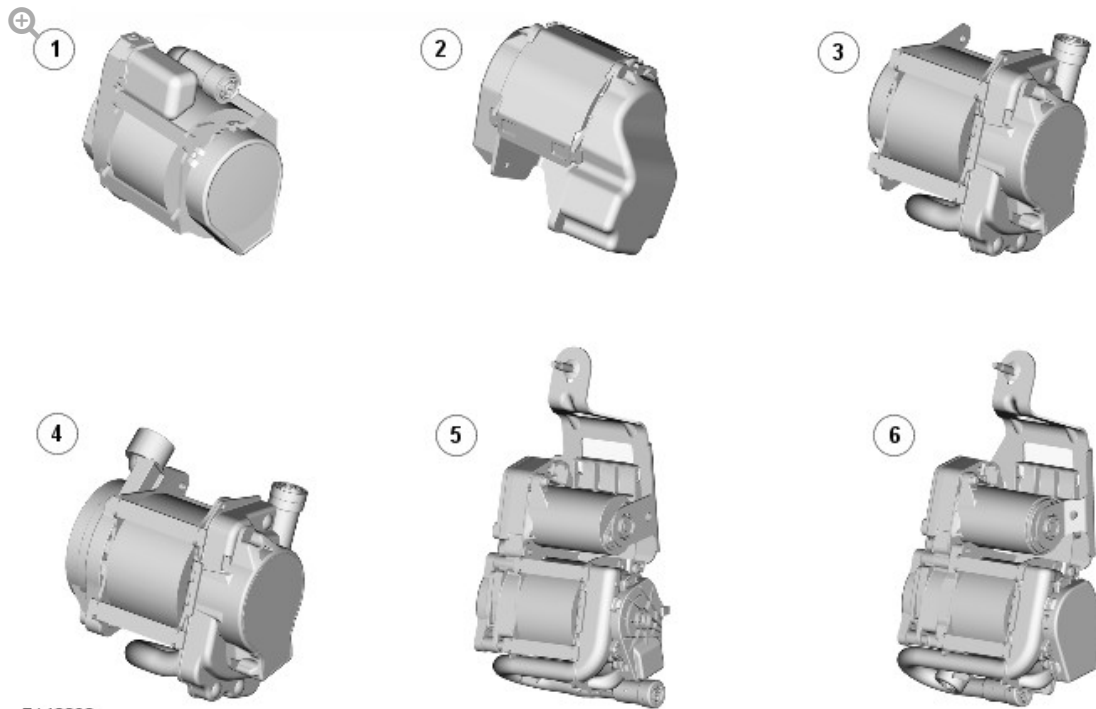
The pretensioners, in the retractors and the driver seatbelt anchor, and the load limiter actuator are pyrotechnic devices that are controlled by the RCM (restraints control module) in the SRS (supplemental restraint system). The tension sensor is part of the occupant classification system in the SRS.

For additional information, refer to: [Airbag and Safety Belt Pretensioner Supplemental Restraint System](#) (501-20B Supplemental Restraint System, Description and Operation).

A seatbelt reminder and a belt minder system warn the driver if the seatbelt of an occupied front seat is not fastened, or if a front or rear seatbelt is unfastened during the journey.

DESCRIPTION

RETRACTORS



E142398

ITEM	DESCRIPTION
1	Outboard rear seatbelt retractor
2	Center rear seatbelt retractor
3	Front seatbelt pretensioner retractor - non-active seatbelt (all except NAS passenger)
4	Front seatbelt pretensioner retractor - non-active seatbelt (NAS passenger only)
5	Front seatbelt pretensioner retractor - active seatbelt (all except NAS passenger)
6	Front seatbelt pretensioner retractor - active seatbelt (NAS passenger only)

All of the seatbelt retractors incorporate an emergency locking function and, except for the center rear seatbelt on 5 seat vehicles, a load limiter function. On NAS vehicles, the retractors of the front passenger and rear seatbelts also incorporate an automatic locking function.

The emergency locking function locks the retractor if the seatbelt is pulled

out of the retractor too quickly, or if the vehicle is subjected to a sudden deceleration or severe tilt angle. The automatic locking function allows the seatbelt to be tightened and locked, for the installation of a child seat.

The load limiter function progressively releases an amount of seatbelt when the belt tension exceeds a certain level, to limit the chest loading of an occupant during an impact.

LOAD LIMITER ACTUATOR

On NAS vehicles, the retractor of the front passenger seatbelt incorporates an adaptive load limiter that has two different load limits, the standard load limit and a lower load limit. The lower load limit is designed to reduce the seatbelt chest loading on smaller seat occupants during an impact. The load limiter changes from the standard limit to the lower limit when the load limiter actuator is fired by the RCM.

For additional information, refer to: [Airbag and Safety Belt Pretensioner Supplemental Restraint System](#) (501-20B Supplemental Restraint System, Description and Operation).

ACTIVE SEATBELTS

Where fitted, active seatbelts incorporate retractors that recognize when the vehicle is in an extreme cornering or braking condition, and automatically retract the seatbelt to pull the occupants more securely into their seats. Each active seatbelt retractor has a locking unit, a drive unit with an electric motor, a clutch and gearing system, and an electronic control module. The control module controls the operation of the electric motor using stored algorithms. The active seatbelt retractor is a non-serviceable unit.

Active seatbelt retractors are powered by an ignition feed from the ignition relay in the CJB (central junction box), and a permanent battery feed from the RJB (rear junction box). High speed CAN (controller area network) chassis system connections provide the electronic control modules in the active seatbelt retractors with vehicle dynamic behavior from other system

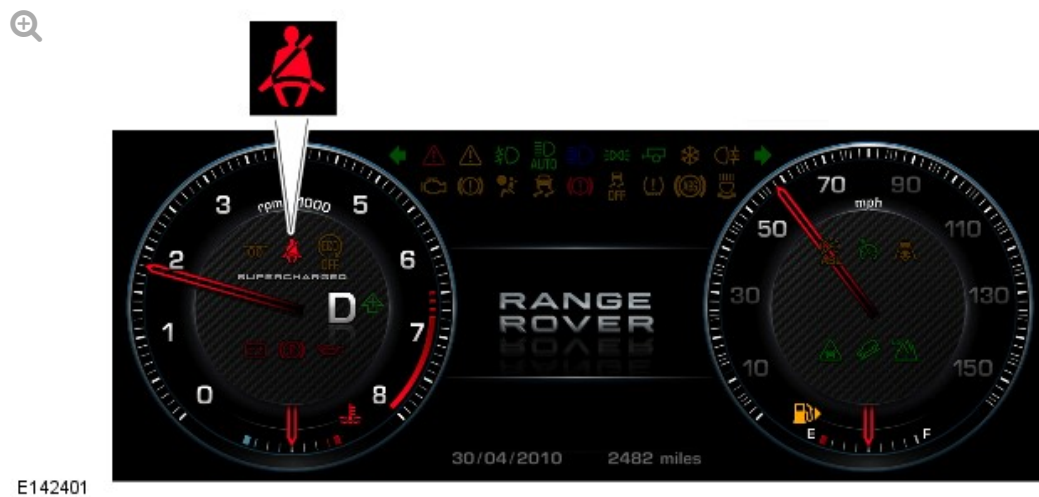
control modules.

BUCKLE SWITCHES

The buckle of each seatbelt incorporates a switch to provide a seatbelt status signal, which is used for operation of the seatbelt reminder and belt minder systems, and for the firing strategy of the SRS.

The switches of the front seatbelts have signal and ground connections with the RCM. The switches of the rear seatbelts have connections to a vehicle ground point and to the CJB. The RCM and the CJB broadcast the status of the switches on the high speed CAN buses.

SEATBELT WARNING INDICATOR



The CJB controls the operation of the seatbelt warning indicator with a high speed CAN powertrain systems message to the instrument cluster.

OPERATION

SEATBELT REMINDER

The seatbelt warning indicator comes on if the driver seatbelt, or the seatbelt of an occupied front passenger seat, is not fastened. The indicator goes off when the offending seatbelt is fastened.

Front passenger seat occupancy status is provide by a hardwired input from the occupant detection sensor (all except NAS), or a high speed CAN

powertrain systems message from the occupant classification sensor control module (NAS only).

BELTMINDER SYSTEM

Where fitted, the belt minder system augments the seatbelt reminder. At vehicle speeds above 10 mph (16 km/h), if the seatbelt of the driver seat or an occupied front passenger seat has not been fastened, or if a front or rear seatbelt is unfastened, the belt minder system activates an intermittent warning tone, flashes the seatbelt warning indicator and displays a warning message indicating which seatbelt triggered the warning.

The CJB controls the operation of the belt minder system with a high speed CAN powertrain systems message to the instrument cluster.

A graphic in the message center shows the status of the seatbelts. Each seating position is represented by a passenger icon, the color of which indicates the seatbelt status:

- No color - seatbelt not fastened.
- Green - seatbelt fastened.
- Red - seatbelt status changed from fastened to unfastened.

A belt minder warning triggered by one of the rear seatbelts can be cancelled by pressing the information button on the steering wheel.

Although not advisable, it is possible to disable the belt minder system using Land Rover approved diagnostic equipment.

NOTE:

If a heavy object is placed on the front passenger seat, it may trigger a belt minder warning.

ACTIVE SEATBELTS

The active seatbelt function is scenario based, with arbitration taking place

between the active scenarios to determine the required system response. The system covers the following four scenarios:

- CMbB (collision mitigation by braking) and AEBA (advanced emergency brake assist), associated with the forward alert function of the adaptive speed control system.
- Braking.
- Cornering.
- Stability (oversteer/understeer).

The active seatbelt system recognizes activation of the AEBA function, heavy braking, hard cornering and the onset of either understeer or oversteer, as unusual states. The system receives the unusual state signals from several network modules which broadcast the information on the high speed CAN buses.

The signals are detected by each retractor's integrated control module, compared against the algorithm database and, if appropriate, the module instructs the electric motor to activate the retractor.

If the unusual state continues to develop into an accident, the seatbelt pretensioners and load limiters are activated in the normal way, over and above the function of the retractor. If the unusual state returns to a usual state, the control module sends a further signal to the electric motor and the retractor is released until required again.

Conditions which activate the system include high lateral cornering forces, exceeding pre-determined understeer or oversteer thresholds and exceeding pre-determined brake pressure thresholds. In each case the force and speed of the seatbelt retraction will be proportional to the severity of the manoeuvre.

The system activates the retractor at lower thresholds with correspondingly lower forces and on-set rates, while increasing both force and the on-set rate in proportion to more severe inputs.

The system is also used to alert the driver to the AEBA function. If the threat of a collision increases beyond a set threshold, the system will retract the seatbelt.

TENSION PROFILES

The operation of the system is linear and progressive in nature. Low level tensioning is introduced as the vehicle enters a manoeuvre that is deemed to offer a customer benefit by tensioning the seatbelt, thus providing occupant support. If the vehicle increases the severity of its manoeuvre, the level of tension will increase, until a pre-determined high level combination of force and speed is reached, at which point 'maximum' output tension will be a response to a critical driving manoeuvre.

Once tension in the seatbelt is achieved (and the seatbelt has stopped pulling in), further increases in manoeuvre severity will not result in an increased tension. This is to prevent the occupant feeling repeated 'jerking' of the seatbelt through a manoeuvre. However, if a critical driving manoeuvre is detected this logic is overruled and the retractor will attempt to pull in at maximum force and speed.

NETWORK SIGNALS

The active seatbelt control modules use a combination of information from other system modules. Using this information the module applies algorithms to control the active seatbelt operating conditions.

The active seatbelt control modules receive signals on the high speed CAN chassis systems from the following system components:

SIGNAL	SIGNAL SUPPLIER
Brake pressure	ABS (anti-lock brake system) control module
Wheel speed sensors	ABS control module
Steering wheel angle	ABS control module
Steering wheel angle speed	ABS control module

Steering wheel angle status	ABS control module
Lateral acceleration	RCM
Yaw rate	RCM
Driver and passenger seatbelt buckle status	RCM
Passenger seat occupant status (NAS only)	RCM
Crash status	RCM
Passenger seat occupant status (all except NAS)	CJB
Vehicle information parameter	CJB
CCF (car configuration files)	CJB
Battery voltage	CJB
Menu setting request	Instrument cluster
Dynamic or Winter mode selected	Transmission control switch
Advanced emergency brake assist	Adaptive speed control module

The active seatbelt control modules communicate with each other to confirm the sensitivity mode request from the instrument pack (selected by the vehicle user) and/or the dynamic or winter mode request from the transmission control switch (selected by the vehicle user).

Acknowledgment signals are broadcast on the high speed CAN chassis systems to confirm active seatbelt status in reply to the input signals listed above. Fault warnings to the instrument pack message center activate an Active Seat Belt Unavailable message, which is accompanied by the triangular amber warning indicator.

DRIVER SETTING

The vehicle user can access the active seatbelt setting menu via the vehicle Set-up menu of the instrument cluster:

- Dynamic Mode On – the active seatbelt system is set to activate during enthusiastic driving and critical driving maneuvers.

- Dynamic Mode Off – the active seatbelt system is set to activate only during critical driving maneuvers with a maximum level output.

The user selection remains locked during and after ignition cycling.

WINTER AND DYNAMIC MODES

The selection of the transmission dynamic mode or winter mode activates alternative mapping in the active seatbelt control modules providing different levels of force and speed sensitivity of the system.

If the active seatbelt sensitivity is set to Dynamic Mode On (as selected in the instrument cluster menu) and the vehicle user selects transmission dynamic mode, the active seatbelt will tension gently and then release. This will occur at all vehicle speeds unless the active seatbelt is inhibited or Dynamic Mode Off is selected.

INHIBITED AND DISABLED FUNCTIONS

The following references to active seatbelt inhibited or disabled are defined as follows:

- Active seatbelt has been inhibited but the control module is operating normally. For example, all scenarios are inhibited because the seatbelt is not fastened.
- Active seatbelt has been disabled due to an internal or external fault. For example, active seatbelt control module internal memory fault or a fault discovered with the ABS module information.

The relevant active seatbelt control module is inhibited if any of the normal operating conditions listed below occurs during a given ignition cycle. As soon as the condition no longer applies in the same ignition cycle, the active seatbelt control module is re-enabled:

- RCM indicates that the driver or passenger seatbelt is not fastened for the relevant active seatbelt control module.

- Ignition is off and no messages are being received or transmitted on the high speed CAN chassis systems as a result.
- NAS vehicles only - the active seatbelt functionality is inhibited on the passenger side if the occupant classification sensor control module indicates that the passenger seat status is occupied inhibit. For example, a small person is occupying the passenger seat.
- RCM is transmitting a crash signal before the start of any retraction.
- Active seatbelt is in the middle of a retraction or is already holding a retraction and the RCM has indicated that a crash has occurred.

NOTE:

No DTC (diagnostic trouble code) is recorded during the inhibited condition.

The relevant active seatbelt control module is disabled if any of the fault conditions listed below occur during a given ignition cycle:

- A fault occurs in high speed CAN message transmission or other system components transmitting operating information.
- Failure of critical parts within the active seatbelt unit, including software, that will not be resolved during a given ignition cycle.
- Battery voltage below 9 V or above 16 V continuously for two seconds (the system will differentiate between genuine low battery voltage and low voltage due to engine cranking).
- NAS vehicles only - the functionality is disabled on the passenger side only because the occupant classification sensor control module indicates a fault.
- The active seatbelt motor has overheated and no retractions are available in the current cycle.
- The active seatbelt unit has exceeded its cycle requirements for lower

functions and safety activation is unavailable (see below).

NOTE:

If the active seatbelt motor exceeds a certain heat threshold and is on a hold cycle, the tension is still released when the manoeuvre has been completed.

The active seatbelt has been designed to exceed 10 years operating life in normal driving conditions. However, it has a limited life cycle. The lower output functions are disabled in the unlikely event that the unit exceeds its cycle requirement due to repeated overuse. In this case the lower output retractions are not operational. If this condition occurs the unit only operates with a maximum output and a DTC is recorded. The DTC remains active even if a complete re-programming procedure of the active seatbelt control module is attempted. The maximum output will operate in all activation scenarios under critical maneuvers previously described except for cornering, which does not include a maximum output activation.

DIAGNOSTICS

In the event of a system fault DTC's are stored in the control modules. In certain circumstances the system will attempt to maintain an operational state utilizing the most appropriate default strategy, reducing its functionality depending on the type of fault. The message Active Seat Belt Unavailable is displayed in the instrument cluster message center, accompanied by the triangular amber warning indicator.

The active seatbelt control modules contain a self-test function, which can be activated by Land Rover approved diagnostic equipment.

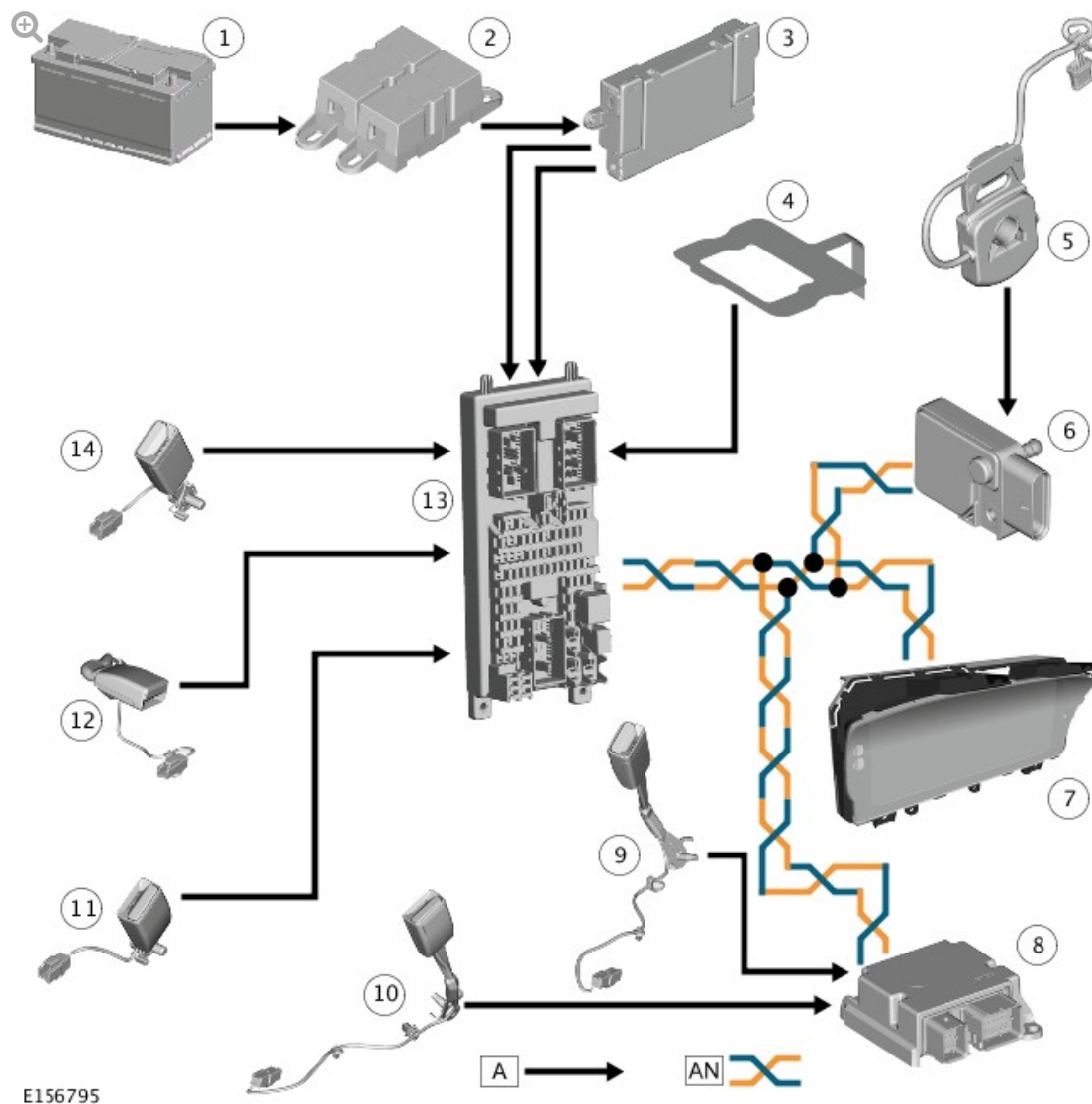
The self-test operates in the following sequence:

- 1** With the seatbelt buckled and the engine running, a maximum retraction is implemented.

2 Retraction held for 3 seconds.

3 Retraction released.

CONTROL DIAGRAM - SEATBELT REMINDER/BELTMINDER SYSTEM

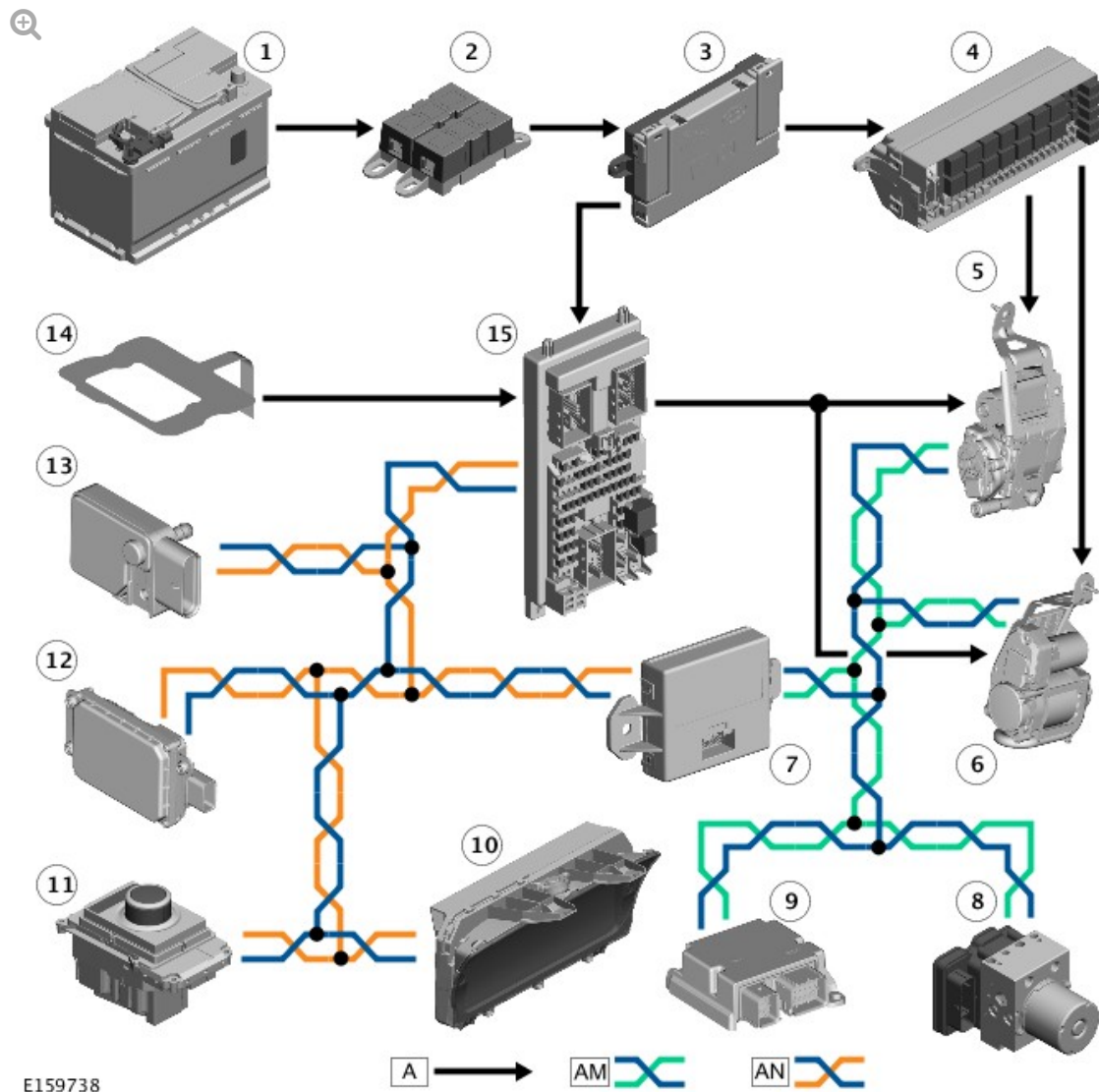


A = HARDWIRED; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Occupant detection sensor (where fitted)
5	Seatbelt tension sensor (where fitted)

6	Occupant classification sensor control module (where fitted)
7	Instrument cluster
8	Restraints control module
9	Left front buckle switch
10	Right front buckle switch
11	Right rear buckle switch
12	Center rear buckle switch (where fitted)
13	Left rear buckle switch
14	Central junction box

CONTROL DIAGRAM - ACTIVE SEATBELTS



E159738

A = HARDWIRED; AM = HIGH SPEED CAN CHASSIS SYSTEMS; AN = HIGH SPEED CAN POWERTRAIN SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	Battery junction box 2
3	Battery junction box
4	Rear junction box
5	Left seatbelt retractor
6	Right seatbelt retractor
7	Gateway module
8	Anti-lock brake system control module
9	Restraints control module
10	Instrument cluster
11	Transmission control switch (dynamic and winter mode switches)
12	Adaptive cruise control module
13	Occupant classification sensor control module (where fitted)
14	Occupant detection sensor (where fitted)
15	Central junction box